many companies have started the fabrication process of 20cm wafers. Some have already demonstrated 20cm SiC samples. These include Cree (now Wolfspeed), II-VI, SiCrystal (a subsidiary of ROHM Group), and STMicroelectronics.

In July 2021, STMicroelectronics announced that it has manufactured the first 20cm SiC bulk wafers for prototyping next-generation power devices. "The transition to 200mm SiC wafers will bring substantial advantages to our automotive and industrial customers as they accelerate the transition towards electrification of their systems and products," says Marco Monti, President Automotive and Discrete Group, STMicroelectronics.

ST's 20cm SiC wafer

5G and SiC. Due to the advent of 5G, there will be demand for high-frequency RF power amplifiers. This is another potential application industry for SiC devices apart from the automotive sector, since GaN (gallium nitride) on SiC is a key technology for 5G power amplifiers. II-IV announced in 2019 that it has introduced prototype 20cm diameter semi-insulating SiC substrates for RF power amplifiers in 5G wireless base-station antennas and other high-performance RF applications.

In-house production. The demand for 20cm SiC is growing and many companies would prefer to produce it in-house. This is why they are expanding their fabrication facilities for the mass production of 20cm wafers. A few years down the line, we can expect several key players in the semiconductor business to have 20cm wafer production facilities.

Bosch aims to become a global leader in the production of SiC chips for electromobility and plans to develop SiC devices on 20cm wafers instead of standard 15cm ones. The company has expanded its wafer fab facilities and is expanding even further for complete in-house production of SiC based devices.

Infineon has been involved in SiC technology for a long time and by 2023 aims to start mass production of 20cm SiC. Cree (Wolfspeed) plans to use existing facilities and install refurbished tools to produce 20cm SiCs in a fiscally responsible manner.

Investments. Another important trend in this industry is the increase in investments in SiC technology. In 2019, Cree (Wolfspeed) announced their largest-ever investment of $1 billion to expand SiC capacity. Moreover, collaborations and acquisitions among semiconductor companies have also become noticeable. Multiple players from all over the world are joining the SiC race and helping the technology become mainstream.

Exciting collaborations. Many semiconductor companies have collaborated, started joint ventures and partnerships in the SiC domain. On 1st February 2022, II-VI Incorporated announced that it has expanded its relationship with GE by signing a three-year technology access agreement (TAA) with GE Research. In November 2021, Japanese power electronics company ROHM and a China based firm known as Zhenghai Group announced a new joint venture that focuses on developing SiC power modules for automotive applications. Many key players in this business have invested in such partnerships with each other.

The future of SiC technology

The competition in the production of 20cm SiC is expected to grow, since the demand for SiC is slowly growing more than the supply. As the automobile industry pushes SiC into the mainstream, most challenges related to 20cm wafers mentioned in this article can be eliminated. However, this will take time, and 15cm wafers might dominate the market for a few more years, after which the 20cm tech will take over. And when this happens, device prices will decrease considerably compared to 15cm SiC.

In a 2005 IEEE paper titled Silicon Carbide Power Devices - Current Developments and Potential Applications, authors Peter Friedrichsi and Roland Rupp mention the following about SiC: "The final market penetration will be somehow related to the fantasy and degree of innovation in circuit design groups, since the simple replacement of silicon by SiC alone cannot justify the cost position resulting from the implementation of SiC."

Today, not only are SiC devices already in the market, we also have justifiable reasons as to why 20cm SiC wafer might very well become as mainstream in the coming years as 20cm silicon wafer is today. EFY

The author, Aaryaa Padhyegurjar, is an Industry 4.0 enthusiast with a keen interest in innovation and research.